Water Supply & Fleet Billing Management System

# Database Schema

Tables, fields, relationships and indexes

| | |
|---|---|
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# 1. About this document

Target: PostgreSQL 16 or newer. This is a logical specification, so tables, constraints and indexes are described rather than scripted.

### Conventions

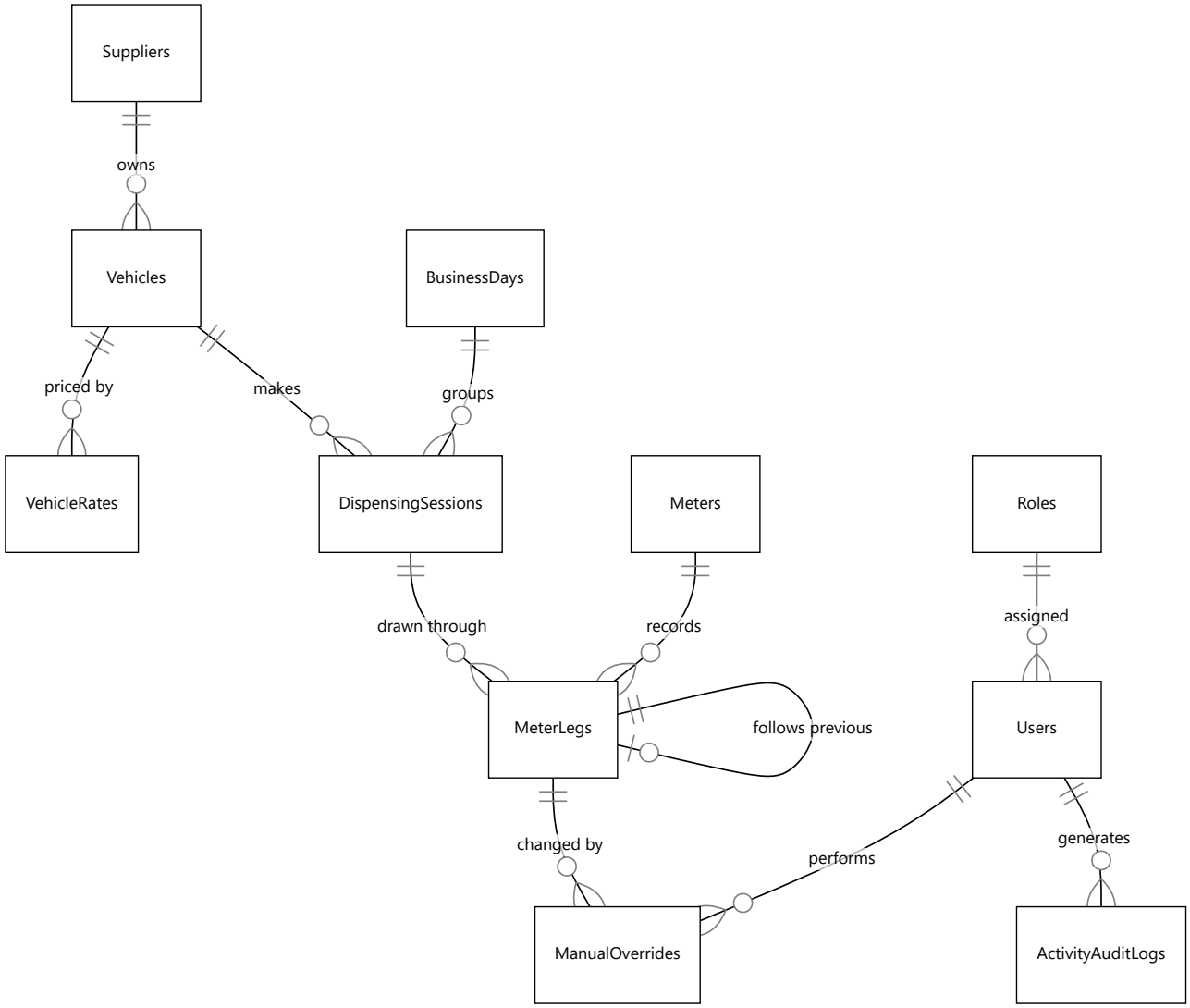
| Topic | Convention |
|---|---|
| Keys | `bigint` identity unless stated |
| Money | `numeric(14,2)`. Never floating point. |
| Readings, volume | `numeric(12,1)` |
| Time | `timestamptz`, stored in UTC, displayed in IST |
| Business dates | `date` |
| Enums | Small lookup or check-constrained text, listed in each table |
| Audit columns on all master tables | `created_at`, `created_by`, `updated_at`, `updated_by`, `version` (for edit conflict checks) |

PK primary key, FK foreign key, UQ unique, NN not null.

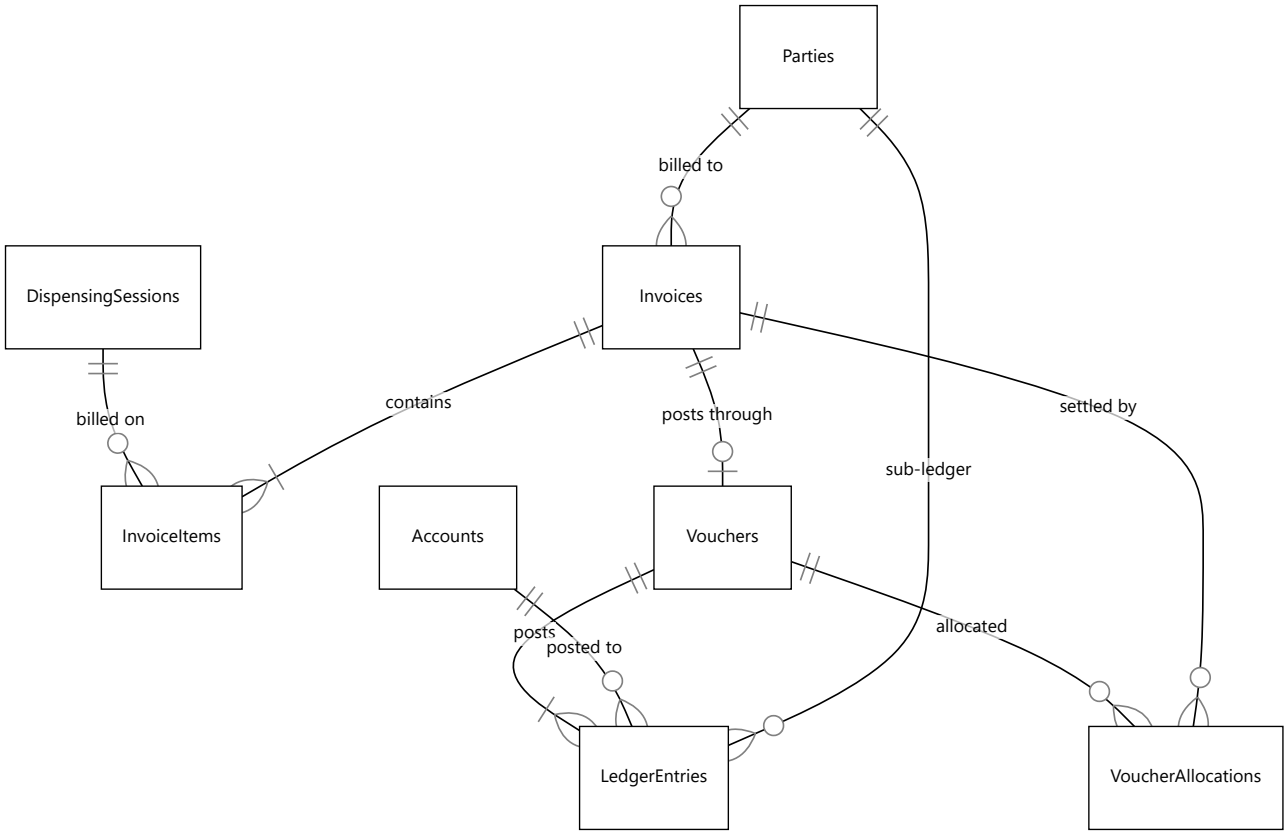
# 2. How the data connects

The data falls into two groups. The first records what happens at the meters. The second records the money. Invoices link the two.

### Operations



**Finance**




Here, Parties means Suppliers, Customers and Staff. Ledger entries also carry a vehicle tag, which produces the vehicle statements.

# 3. Main tables

### `Suppliers`

| Field | Type | Rules | Screen label |
|---|---|---|---|
| `supplier_id` | bigint | PK | |
| `supplier_name` | varchar(120) | NN, UQ ignoring case, spaces tidied | Supplier Name* |
| `short_name` | varchar(30) | NN | Used for nickname |
| `contact_person` | varchar(80) | | Contact Person |
| `phone_no` | varchar(15) | NN | Phone No.* |
| `alt_phone_no` | varchar(15) | | Alternate Phone No. |
| `address` | text | | Address |
| `opening_balance` | numeric(14,2) | Default 0 | |
| `opening_balance_type` | char(2) | `Dr` or `Cr` | |

| Field | Type | Rules | Screen label |
|---|---|---|---|
| status | varchar(10) | NN, ACTIVE or INACTIVE | Status |

**Indexes:** unique on lower-case name; status.

## Vehicles

| Field | Type | Rules | Screen label |
|---|---|---|---|
| vehicle_id | bigint | PK | |
| vehicle_code | varchar(10) | NN, UQ, never reused | Vehicle Code (invoice "Vehicle No.") |
| vehicle_no | varchar(20) | NN, UQ, uppercase | Vehicle No.* |
| nickname | varchar(80) | NN | Vehicle Name |
| supplier_id | bigint | NN, FK Suppliers | Supplier* |
| tank_capacity_litres | integer | NN, above 0 | Capacity (Litres)* |
| billing_basis | varchar(10) | NN, TRIP or VOLUME, default TRIP | |
| trip_cost | numeric(10,2) | NN, 0 or more. No ledger effect | Trip Cost (₹)* |
| unit_of_measure | varchar(10) | NN, default TRIP | Unit of Measurement |
| water_details | varchar(120) | | Water Details |
| driver_name | varchar(80) | | Driver details |
| driver_phone | varchar(15) | | Driver details |
| driver_licence_no | varchar(25) | | Driver details |
| status | varchar(10) | NN, ACTIVE, INACTIVE or MAINTENANCE | Vehicle Status* |

**Indexes:** unique vehicle_code; unique vehicle_no; supplier_id, status; trigram index on vehicle_no and nickname for search.

## Meters

| Field | Type | Rules |
|---|---|---|
| meter_id | smallint | PK, only 1, 2 or 3 |
| meter_name | varchar(20) | NN, UQ |
| unit | varchar(5) | NN, default kL |
| max_reading | numeric(12,1) | NN, default 99999.9 |
| display_step | numeric(4,1) | NN, default 0.1 |
| current_reading | numeric(12,1) | NN. Latest end reading in the chain |
| last_chain_end_at | timestamptz | NN |
| last_leg_id | bigint | FK MeterLegs |
| chain_version | integer | NN, default 1. Rises on chain reset |

| Field | Type | Rules |
|---|---|---|
| status | varchar(10) | NN, IDLE, FILLING, FAULT or OFFLINE |

The row check limiting IDs to 1, 2 and 3 enforces "exactly three meters". Each start and stop takes a row lock on this table, which serialises one meter while leaving the others untouched.

### DispensingSessions

One vehicle visit.

| Field | Type | Rules |
|---|---|---|
| session_id | bigint | PK |
| vehicle_id | bigint | NN, FK Vehicles |
| supplier_id | bigint | NN, FK Suppliers. Copied at start, so later transfers do not rewrite history |
| business_date | date | NN, FK BusinessDays |
| in_time | timestamptz | NN |
| out_time | timestamptz | Null while open |
| total_volume | numeric(12,1) | NN, default 0. Sum of active legs |
| capacity_warning_reason | text | Required if over tolerance |
| billing_intent | varchar(10) | NN, NOW or LATER |
| status | varchar(12) | NN, OPEN, COMPLETED, UNBILLED, INVOICED, ABORTED, WRITTEN_OFF |
| written_off_reason | text | Required when written off |
| is_manual | boolean | NN, default false. Never reset |
| version | integer | NN, edit conflict check |

**Unique rule:** one row per vehicle where status is OPEN. **Indexes:** business_date, status; vehicle_id, in_time; supplier_id, business_date; partial on status for COMPLETED and UNBILLED.

### MeterLegs

One vehicle's draw on one meter. This table is the meter chain.

| Field | Type | Rules |
|---|---|---|
| leg_id | bigint | PK |
| meter_id | smallint | NN, FK Meters |
| chain_version | integer | NN |
| session_id | bigint | FK DispensingSessions. Null only for an unresolved unattributed leg |
| prev_leg_id | bigint | FK MeterLegs, UQ. Null only on the first leg of a chain version |

| Field | Type | Rules |
|---|---|---|
| leg_type | varchar(14) | NN, NORMAL, UNATTRIBUTED or ABORTED |
| start_reading | numeric(12,1) | NN. Copied from previous end, not user input |
| end_reading | numeric(12,1) | Null while open |
| rollover_applied | boolean | NN, default false |
| volume | numeric(12,1) | Set on stop, 0 or more |
| chain_start_at | timestamptz | NN. Copied from previous chain end |
| chain_end_at | timestamptz | Null while open |
| actual_start_at | timestamptz | NN. Real start, used for In Time |
| actual_end_at | timestamptz | Real stop, used for Out Time |
| capture_source | varchar(16) | NN, DEVICE, OPERATOR_LIVE or MANUAL_OVERRIDE |
| resolved_by, resolved_at, resolution_note | | For unattributed legs |

### How the database enforces the chain:

| Rule | Mechanism |
|---|---|
| One open leg per meter | Unique on meter_id where end_reading is null and type is not ABORTED |
| No two legs share a predecessor | Unique on prev_leg_id |
| No time overlap on a meter | Exclusion constraint on meter_id, chain_version and the chain time range |
| No reading gaps | Check on insert: start_reading must equal the predecessor's end_reading |
| A leg is either open or fully closed | Check: end_reading, chain_end_at and volume are all null or all present |

**Indexes:** meter_id, chain_version, chain_start_at; session_id; partial on leg_type for UNATTRIBUTED.

## Invoices

| Field | Type | Rules | Screen label |
|---|---|---|---|
| invoice_id | bigint | PK | |
| invoice_no | bigint | NN, UQ, gapless, from NumberSeries | Invoice No. |
| invoice_type | varchar(14) | NN, TRIP or CONSOLIDATED | |
| invoice_date | date | NN | Invoice Date |
| business_date | date | NN, FK BusinessDays | Value Date |
| bill_to_type | varchar(10) | NN, SUPPLIER or CUSTOMER | |
| supplier_id | bigint | FK Suppliers | Vehicle Supplier* |
| customer_id | bigint | FK Customers | Bill To |

| Field | Type | Rules | Screen label |
|---|---|---|---|
| payment_mode | varchar(10) | NN, CASH or CREDIT | Payment Mode* |
| total_trips | integer | NN | Trip* |
| total_volume | numeric(12,1) | NN | Meter total |
| total_amount | numeric(14,2) | NN, 0 or more | Total (₹) |
| allocated_amount | numeric(14,2) | NN, default 0 | |
| narration | text | | Narration |
| status | varchar(12) | NN, DRAFT, POSTED, PART_PAID, PAID or CANCELLED | |
| voucher_id | bigint | FK Vouchers, UQ | |
| cancelled_by, cancelled_at, cancel_reason | | Required when cancelled | |
| replaces_invoice_id | bigint | FK Invoices | Correction link |

**Check:** exactly one of supplier_id or customer_id, matching bill_to_type. **Indexes:** unique invoice_no; business_date, status; supplier_id, business_date; customer_id, business_date; partial on open balances (POSTED, PART_PAID) for ageing.

## InvoiceItems

| Field | Type | Rules | Screen label |
|---|---|---|---|
| item_id | bigint | PK | |
| invoice_id | bigint | NN, FK Invoices | |
| line_no | integer | NN, UQ with invoice_id | S.No |
| session_id | bigint | FK DispensingSessions | |
| vehicle_id | bigint | NN, FK Vehicles | |
| material_description | varchar(120) | NN | Material |
| quantity | numeric(12,1) | NN, above 0 | Qty |
| measurement | varchar(10) | NN | Measurement |
| rate | numeric(10,2) | NN. Copied at save, never re-derived | Rate (₹) |
| rate_source | varchar(10) | NN, MASTER or OVERRIDDEN | |
| amount | numeric(14,2) | NN | Total Amount (₹) |
| volume | numeric(12,1) | NN | |
| is_active | boolean | NN, default true. False when invoice cancelled | |

**Unique rule:** session_id where is_active is true, so a session cannot be billed twice (FR-INV-09). **Indexes:** invoice_id; vehicle_id, invoice_id; session_id.

## Vouchers

The header for every posting: receipts, payments, sales, closing adjustments, opening entries and reversals.

| Field | Type | Rules | Screen label |
|---|---|---|---|
| voucher_id | bigint | PK | |
| voucher_type | varchar(12) | NN, SALES, RECEIPT, PAYMENT, JOURNAL, REVERSAL | |
| voucher_no | bigint | NN | Receipt No. / Voucher No. |
| display_no | varchar(20) | NN, UQ, e.g. RV-432, PV-432, INV-7232 | Voucher / Reference No |
| voucher_date | date | NN | Date / Payment Date |
| business_date | date | NN, FK BusinessDays | Value Date |
| party_type | varchar(10) | SUPPLIER, CUSTOMER, STAFF or null | Party / Supplier* |
| supplier_id, customer_id, staff_id | bigint | FKs. At most one set, matching party_type | Pay To |
| vehicle_id | bigint | FK Vehicles | |
| purpose_account_id | bigint | FK Accounts | Ledger* (Payment Purpose) |
| mode | varchar(6) | CASH, BANK, UPI or null | Mode of Receipt / Payment Mode |
| reference_no | varchar(40) | Required for Bank and UPI | Reference Number |
| amount | numeric(14,2) | NN, above 0 | Amount* |
| narration | text | | Narration |
| status | varchar(10) | NN, POSTED or CANCELLED | |
| reverses_voucher_id | bigint | FK Vouchers, UQ | |
| cancel_reason | text | | |

**Unique rule:** voucher_type, voucher_no. **Indexes:** business_date, voucher_type; supplier_id, voucher_date; customer_id, voucher_date; staff_id, voucher_date; reference_no.

## LedgerEntries

Append-only. The application's database account has no update or delete right here.

| Field | Type | Rules | Statement column |
|---|---|---|---|
| entry_id | bigint | PK | |
| posting_seq | bigint | NN, UQ, rises with each line | Sort order |
| voucher_id | bigint | NN, FK Vouchers | |
| line_no | integer | NN, UQ with voucher_id | |
| account_id | bigint | NN, FK Accounts | |
| party_type | varchar(10) | | |

| Field | Type | Rules | Statement column |
|---|---|---|---|
| supplier_id, customer_id, staff_id | bigint | FKs. At most one set | |
| vehicle_id | bigint | FK Vehicles | |
| entry_date | date | NN | Date |
| value_date | date | NN | Value Date |
| particulars | varchar(200) | NN | Particulars / Description |
| reference_no | varchar(40) | NN | Voucher / Reference No |
| debit | numeric(14,2) | NN, default 0 | Debit (Dr) |
| credit | numeric(14,2) | NN, default 0 | Credit (Cr) |
| running_balance | numeric(16,2) | NN, signed: Dr positive, Cr negative. Scope is account + party | Running Balance |
| vehicle_running_balance | numeric(16,2) | Scope is account + party + vehicle. Null if no vehicle | Vehicle statement balance |
| reverses_entry_id | bigint | FK LedgerEntries, UQ | |
| created_by | bigint | NN, FK Users | |
| created_at | timestamptz | NN | |

**Checks:** exactly one of debit or credit is above zero; at most one party set. A deferred check at transaction end confirms that each voucher's debits equal its credits.

### Indexes

| Index | Used by |
|---|---|
| account_id, supplier_id, posting_seq | Supplier statement |
| account_id, customer_id, posting_seq | Customer statement |
| account_id, staff_id, posting_seq | Staff statement |
| vehicle_id, posting_seq | Vehicle statement |
| account_id, entry_date, posting_seq | Account and cash book statements |
| value_date, account_id | Day and period reports |
| voucher_id | Drill-down |

Yearly range partitioning on `entry_date` can be added if the table grows past tens of millions of rows. At the sized volume it is unnecessary.

**How running balance stays correct without rewriting history.** Lines are written in `posting_seq` order under a lock on the balance row, and each stores the balance at that moment. A late correction posts as a new line with today's Date and the original Value Date, so no earlier line changes. A statement's opening balance is the `running_balance` of the last line before the start date.

## Users

| Field | Type | Rules |
|---|---|---|
| user_id | bigint | PK |
| username | varchar(50) | NN, UQ ignoring case |
| full_name | varchar(100) | NN |
| email, phone_no | | |
| password_hash | text | NN |
| mfa_secret_encrypted | text | |
| role_id | bigint | NN, FK Roles |
| extra_permissions | text[] | e.g. override.dispensing |
| must_change_password | boolean | NN, default true |
| failed_login_count | smallint | NN, default 0 |
| locked_until | timestamptz | |
| last_login_at | timestamptz | |
| status | varchar(10) | NN, ACTIVE or INACTIVE |

**Indexes:** unique lower-case username; role_id.

## ActivityAuditLogs

Append-only. No update or delete rights for any application role.

| Field | Type | Rules |
|---|---|---|
| audit_id | bigint | PK, rises with each event |
| event_at | timestamptz | NN, server time |
| user_id | bigint | FK Users. Null for failed logins with unknown username |
| username_snapshot | varchar(50) | NN |
| role_snapshot | varchar(30) | |
| action | varchar(50) | NN, e.g. OVERRIDE_APPLIED |
| category | varchar(16) | NN, AUTH, MASTER, DISPENSING, OVERRIDE, BILLING, FINANCE, DAY, ADMIN, EXPORT |
| severity | varchar(8) | NN, INFO, NOTICE or CRITICAL |
| entity_type, entity_id | | |
| before_value, after_value | jsonb | |
| reason_code, reason_text | | |
| is_post_cutoff | boolean | NN. True for actions after 6 PM |

| Field | Type | Rules |
|---|---|---|
| ip_address | inet | |
| device_info | varchar(200) | |
| request_id | uuid | Links related events |
| prev_hash, entry_hash | char(64) | NN. SHA-256 chain |

**Indexes:** event_at; user_id, event_at; entity_type, entity_id; category, event_at; partial on severity for CRITICAL. Use BRIN on event_at once large.

# 4. Supporting tables

| Table | Key fields | Purpose |
|---|---|---|
| Customers | customer_id PK, customer_name UQ, phone_no, address, opening balance, status | Direct billing parties |
| Staff | staff_id PK, staff_name, phone_no, user_id FK nullable, opening balance, status | Staff Outstanding. Not every worker is a login user |
| VehicleRates | rate_id PK, vehicle_id FK, rate, rate_unit, effective_from, effective_to | Vehicle-wise Rate. Date ranges for one vehicle cannot overlap |
| Accounts | account_id PK, account_name UQ, account_type (ASSET, LIABILITY, EQUITY, INCOME, EXPENSE), is_system, is_party_control, is_cash_like, status | Ledger Master |
| LedgerBalances | balance_id PK, account_id FK, party_type, party_id, vehicle_id, current_balance, last_posting_seq, UQ across scope | Current balance, and the row locked during posting |
| VoucherAllocations | allocation_id PK, voucher_id FK, invoice_id FK, vehicle_id FK, amount, is_active | Links receipts to invoices. Active total cannot exceed either side |
| ManualOverrides | override_id PK, leg_id or session_id FK, user_id FK, reason_code, reason_note, before_value, after_value, impact_summary, applied_at, reversal_voucher_id FK, corrected_voucher_id FK | One row per override |
| MeterChainResets | reset_id PK, meter_id FK, old_final_reading, new_start_reading, reason, reset_by, reset_at | Replacement and recalibration |
| BusinessDays | business_date PK, window_start, window_end, status (OPEN, CLOSED), opening_cash, expected_closing_cash, actual_cash, cash_difference, closed_by, closed_at | Day Opening and Day Closing |
| DayReopenings | reopen_id PK, business_date FK, reason, reopened_by, reopened_at | Reopen history |
| NumberSeries | series_code PK (INVOICE, RECEIPT, PAYMENT, JOURNAL), prefix, next_no | Gapless numbers. Row is locked inside the save transaction, so a rollback keeps the number |
| Roles | role_id PK, role_name UQ | Four roles |

| Table | Key fields | Purpose |
|---|---|---|
| RolePermissions | role_id FK, permission_code, composite PK | Permission set |
| SystemSettings | setting_key PK, setting_value, updated_by, updated_at | Cut-off, override window, tolerance |
| IdempotencyKeys | key PK, request_hash, response, created_at | Prevents double submission |

# 5. Starting data

| Table | Seed |
|---|---|
| Meters | Meter 1, Meter 2, Meter 3, with current physical readings at go-live |
| Accounts (system) | Cash in Hand, Bank, UPI Clearing, Trade Receivables – Suppliers, Trade Receivables – Customers, Staff Advances, Water Sales, Cash Over/Short, Opening Balance Equity |
| Accounts (from design) | Vehicle Maintenance, Fuel Expense, Oil Change, Others |
| Roles | Admin, Dispatch Operator, Billing Clerk, Accountant |
| SystemSettings | business_cutoff 18:00, override_window_start 18:00, override_window_end 23:59:59, capacity_tolerance_pct 5 |

# 6. Data safety during saves

| Operation | Locks, in order | Happens as one unit |
|---|---|---|
| Start fill | Meter row(s), lowest ID first | Session, leg(s), meter status, audit |
| Stop fill | Meter row | Leg close, volume, meter reading, session update, audit |
| Save invoice | Number series, balance rows in ascending key order | Invoice, items, voucher, ledger lines, balances, session status, audit |
| Save voucher | Number series, balance rows in ascending key order | Voucher, ledger lines, balances, allocations, audit |
| Apply override | Meter row, session, then number series and balances if needed | Legs, override record, reversal, correction, audit |
| Close day | Business day row | Checks, any cash voucher, lock, next opening, audit |

Fixed lock order means two operations cannot deadlock. Audit log appends serialise briefly on the hash chain's latest row. This has negligible cost at the sized volume.